Project Design Phase
Problem – Solution Fit Template

Date	17 July 2026
Team ID	-----
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	2 Marks

Problem – Solution Fit Template:

Template:

Canvas Section	Content for Your Project
1. CUSTOMER SEGMENT(S) [CS]	• Residents living in flood-prone areas • Farmers • Disaster Management Authorities • Local Government Officials • Environmental Agencies • NGOs involved in disaster relief
2. PROBLEMS / PAINS + ITS FREQUENCY [PR]	• Lack of timely flood prediction before heavy rainfall • Property and crop damage during floods • Delayed emergency response • Financial losses due to unexpected flooding • Frequent during monsoon seasons and extreme weather events
3. TRIGGERS TO ACT [TR]	• Heavy rainfall alerts • Rising river or reservoir water levels • Weather department warnings • Seasonal monsoon forecasts • News of nearby flooding incidents
4. EMOTIONS BEFORE / AFTER [EM]	**Before:** Fear, anxiety, uncertainty, stress, helplessness. **After:** Confidence, preparedness, safety, peace of mind, improved decision-making.
5. AVAILABLE SOLUTIONS – PROS & CONS [AS]	**Existing Solutions:** Weather forecasts, government alerts, flood monitoring websites. **Pros:** Easily accessible, official information. **Cons:** Generalized forecasts, delayed alerts, limited local prediction accuracy, lack of intelligent risk analysis.
6. CUSTOMER LIMITATIONS [CL]	• Limited access to real-time flood prediction tools • Poor internet connectivity in rural areas • Lack of technical knowledge • Limited availability of location-specific flood information • Budget constraints for advanced monitoring systems
7. BEHAVIOR + ITS	• Regularly checking weather updates during monsoon • Monitoring local news

Canvas Section	Content for Your Project
INTENSITY [BE]	and social media • Contacting local authorities during emergencies • High engagement during heavy rainfall periods; moderate engagement during normal weather conditions
8. CHANNELS OF BEHAVIOR [CH]	**Online:** Web application, Weather websites, Government disaster portals, Social media, Mobile browsers. **Offline:** Television, Radio, Local government announcements, Community volunteers, Disaster management offices.
9. PROBLEM ROOT / CAUSE [RC]	• Climate change causing unpredictable rainfall • Inadequate early warning systems • Lack of AI-based localized flood prediction • Dependence on manual analysis • Insufficient integration of historical rainfall data with predictive analytics
10. YOUR SOLUTION [SL]	**Rising Waters** is an AI-powered flood prediction system that uses **XGBoost Machine Learning** to analyze rainfall and meteorological data. The system preprocesses user input, predicts flood risk through a Flask web application, and provides timely, accurate, and location-aware flood predictions. It supports disaster preparedness, helps authorities make informed decisions, and reduces potential loss of life and property through early warning capabilities.